

# SAINAGESH VEERAVALLI

 (470)-807-5170  [nageshv.9832@gmail.com](mailto:nageshv.9832@gmail.com)  [linkedin.com/in/nagesh9832](https://www.linkedin.com/in/nagesh9832)  [github.com/nageshnen](https://github.com/nageshnen)

## EDUCATION

### THE UNIVERSITY OF GEORGIA | 2023

*Master's in Computer Science.*

### K.L University

*Bachelor's in Computer Science Engineering.*

## TECHNICAL SKILLS

**Languages:** Python, Rust, Go, Java.  
**ML/DL Frameworks:** TensorFlow, Pytorch, Keras, Google ADK.  
**ML Tools:** MLflow, Weights & Biases  
**Computer Vision:** OpenCV, FER, YOLO, GAN's.  
**Cloud Technologies:** GCP, AWS.  
**CI/CD:** Docker, Azure Pipeline, GitLab CI/CD.  
**Frameworks:** Django, Flask, Spring, REST api.

## CERTIFICATIONS

Google Cloud Certified Generative AI Leader.  
Google AI Professional Certificate.  
SAS Advanced Programmer.  
ANAPLAN LEVEL-1 MODEL BUILDER.  
CISCO Packet Tracer.  
Google Data Analytics Professional.  
Google IT Support Certificate.

## PROFESSIONAL EXPERIENCE

### The Home Depot

*Software Engineer 2*

**May 2024 - Present**

*Atlanta, GA*

- \* Partnered with business stakeholders to analyze and refine existing processes and payout calculation methodologies, enhancing accuracy and transparency.
- \* Designed and implemented Tableau dashboards, leveraging data from GCP and integrated with Salesforce CRM, to provide actionable insights and performance tracking for the sales team.
- \* Automated weekly reporting processes within Anaplan, reducing manual effort by 8 hours per week and improving reporting timeliness.
- \* Spearheaded a proof-of-concept(PoC) using the Google's ADK to architect and deploy intelligent AI agents that streamlined operations for key business partners.

**Skills:** Python, GCP, Anaplan, Tableau.

### Qverge

*M.L Engineer*

**Mar 2024 - May 2024**

*Alpharetta, GA*

- \* Led a team of 5 in spearheading a Video Sentiment Analysis project, employing Deepface and FER models to decipher emotions and sentiments from video content, contributing to enhanced understanding of user engagement and feedback.
- \* Collaborated on Gas Cylinder Detection, Tracking, and Monitoring project using YOLOV8x model with CVAT for custom data annotation. Employing MLflow for efficient project management.
- \* Implemented Agile methodology with daily standup meetings and facilitated streamlined collaboration by utilizing Confluence and Azure Cloud for model deployment in project workflows.

**Skills:** MLflow, DeepFace, SSD, AWS, HuggingFace, Flask, Docker.

### Pinnacle Leasing and Management

*Software Engineer*

**May 2023 - Feb 2024**

*Alpharetta, GA*

- \* Employed Azure DevOps for ticket management, ensuring efficient issue tracking, and maintained the latest code base with Azure Repo for code version control and collaboration.
- \* Implemented Azure Pipelines for continuous integration and delivery, enhancing development and deployment. Utilized AWS services like S3 and CloudFront for high-performance website hosting.
- \* Automated infrastructure provisioning with Terraform across various web environments for enhanced resource management and scalability, while also creating user-friendly documentation sites with MkDocs.

**Skills:** Terraform, AzureDevops, AWS, SonarQube, Figma, Mkdocks, Postman, ZohoCRM,.

# PROJECTS

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## Gated Recurrent Unit

**GitHub Repo**

*python/scala*

- GRU model consists of input, hidden, and output layers. It uses a set of hyperparameters like batch size, maximum number of epochs, and optimization techniques like RMSProp or Adam for training.
- Implemented the forward and backward pass of the GRU model. The forward pass computes predictions based on input sequences, while the backward pass calculates gradients for weight updates during training.
- Utilized a sample lake level dataset to perform testing and validation of the GRU model. This dataset is used to evaluate the model's performance in forecasting lake levels, demonstrating its practical application.

## 3-D Object Detection and Identification

**GitHub Repo**

*Python*

- Seamlessly aligned high-resolution LiDAR point clouds with camera images from the KITTI dataset, establishing tightly synchronized data pairs crucial for advanced object detection models and cross-modal insights.
- Leveraged Frustum PointNet and Frustum CloudNet to extract rich features and enhance object detection accuracy within the defined regions of interest (frustums).
- Integrated frustum extraction, feature extraction, and visualization to develop a robust system for 3D object detection in complex real-world environments.

## Image Search Engine

**GitHub Repo**

*python*

- Developed a CBIR image search engine, integrating feature extraction and similarity measurement techniques.
- Elevated search precision through pioneering a background-based image retrieval system, utilizing advanced feature-based similarity calculations, including color histograms and texture descriptors
- Evaluated and fine-tuned the system with diverse datasets, showcasing its practical effectiveness.